

Experiment :- Study of LAN Topology, Network devices and address configuration.

Objective :-

- 1) To study the LAN topology used in college campus.
- 2) To identify networking device & understand their function.
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of network for wired & wireless connections.
- 5) To obtain MAC, IP address & subnet mask.

Problem Statement - 1

Draw & identify LAN topology of college campus & clearly label all components such as clearly label all components such as router, switches & etc.

Soln → The college campus network observed this study consists of:

- * Layer-3 core switches in the server room
- * Managed by POET switches.
- * Wireless access point (APs)
- * Ethernet port in classroom & laboratories.

→ Network Topology

Hybrid Hierarchical Topology which combines:

- 1) Tree Topology
- 2) Extended Star Topology
- 3) Ring Topology (Server room)

→ The network hierarchical layer

- ① Core layer
- ② Distribution layer
- ③ Access layer

→ Core Layer

Core layer contains server room and server room have Firewall, Router, 2-L3+ switches, patch Rack, server, inverter, etc.

→ Distribution layer

* Distribution layer contains 4-switches.

* The 4-switches are →

- ① Main Build → R7 (Room no - 213, 2nd Floor)
- ② Main Build → R1 (Room no - 301, 3rd Floor)
- ③ Main Build → R6 (Room no - 113, 1st Floor)
- ④ Main Build → R2 (Room no - 201, 2nd Floor)

* This Four switches contains →

① 18 port Gigabit Easy Smart switch
with 16-port · TL-SG1218MPE

② SG 34 28 x MP → 24 port Gigabit & 4-port
10 G SFP, L2+ managed switches with
24 port PoE.

→ Now Let's discuss these 2 above components

① T P-Link TL-SG1218MPE

- * Likely used for: CCTV camera, wifi access point etc.
- * 18 port Gigabit "Easy Smart" switch.
- * 16 port support PoE

② TP-Link Omada SG 3428 x MP

- * Higher-end managed Layer 2+ switches.
- * 24 PoE port, Uplinks, 10 G SFP
- * Usually acts as a main distribution for each floor

Access Layer

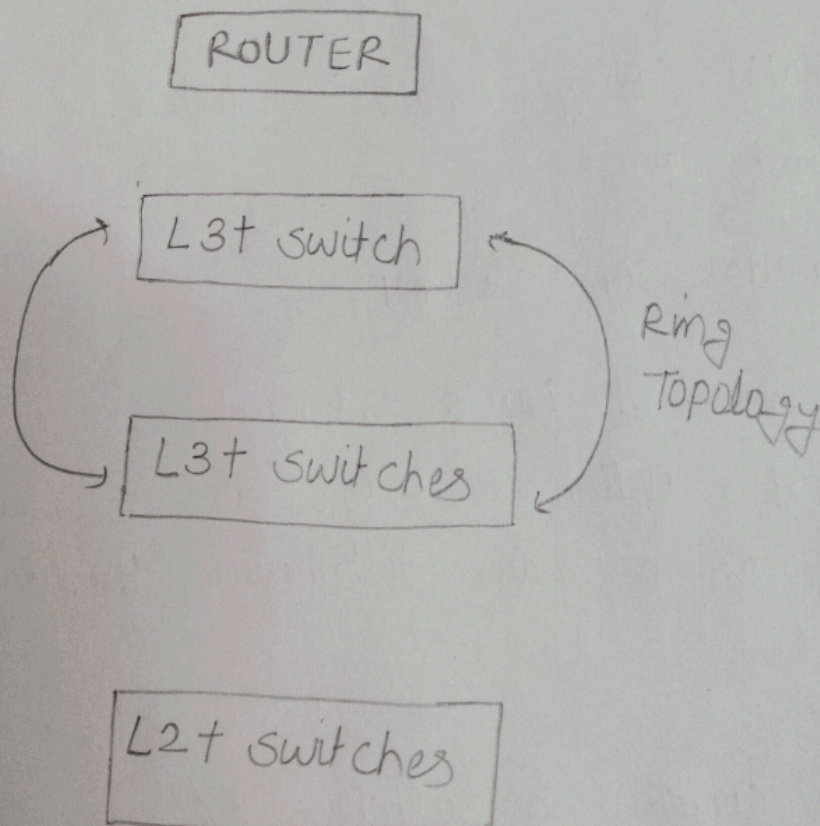
- * Access layer contain 3-switches i.e
 - 1) Main Build R1 - [Room no- 301, 3rd Floor]
 - 2) Main Build R4 [Room no - 05, Ground Floor]
 - 3) Main Build R5 [Room no- 18, Ground Floor]

* Every switches acts as access layer which helps to provide network connectivity to provide network connectivity to all classroom, Lab, Faculty Room, etc.

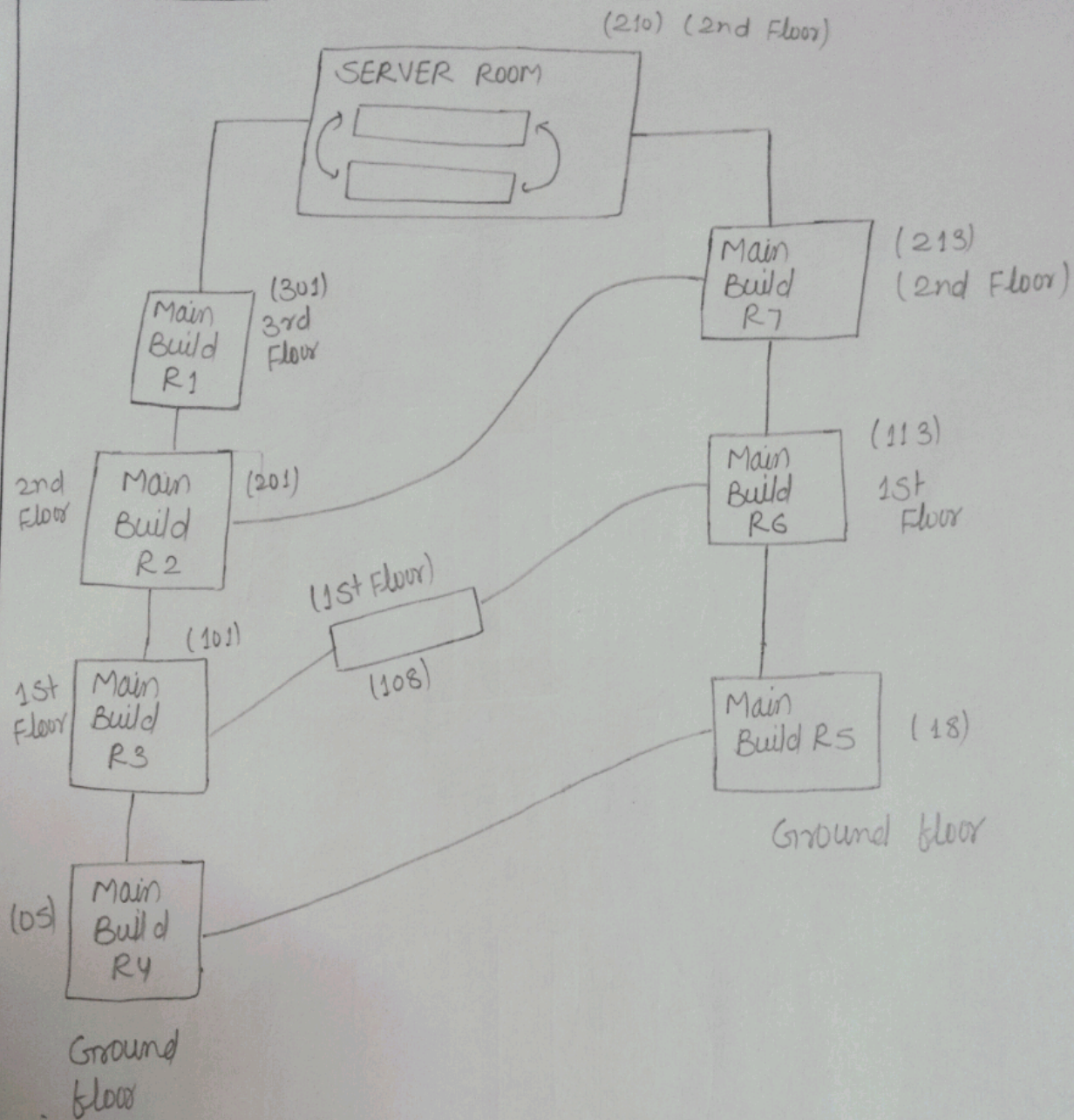
* Access Layer directly connect:

- ① Ethernet port
- ② wireless AP
- ③ Lab System
- ④ outdoor wireless AP (ex - 018)

→ Server Room Architecture / Topology



TOPOLOGY

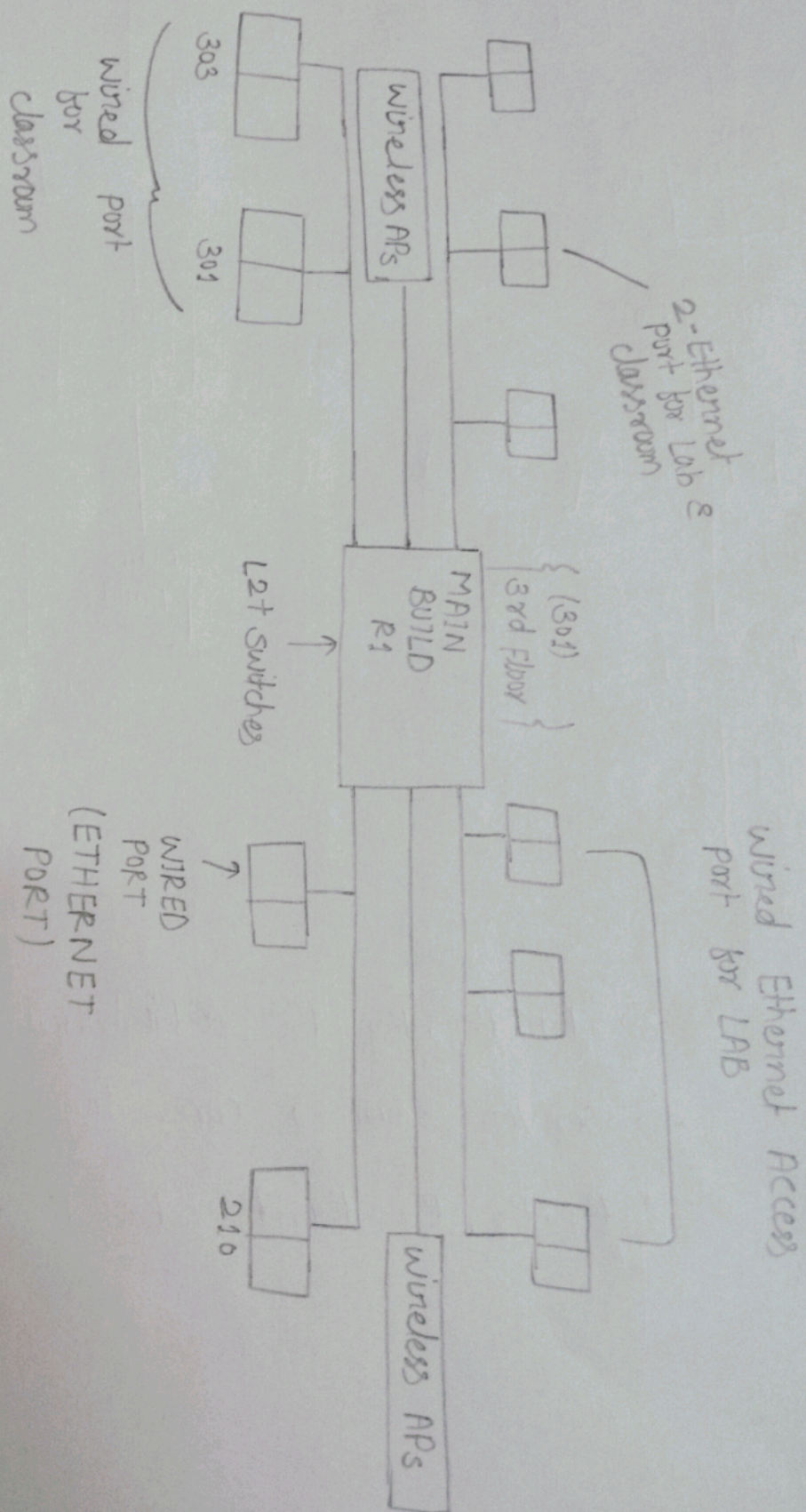


→ R₁, R₇, R₂, R₆ → Distribution layer

→ Server room → Core layer

→ R₁, R₂, R₃, R₄, R₅, R₆, R₇ → Access layer

ACCESS LAYER SWITCH (3rd Floor - 301)



→ Difference between wired & wireless speed?

Solⁿ

Wired Network

- * Faster communication
- * Stable connection
- * Low latency
- * Dedicated physical medium

Wireless Network

- * Slower communication
- * Affected by distance/interference
- * Higher latency
- * Shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q- Role of subnet mask in network ?

Solⁿ

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

It is important for:

- ① Network segment
- ② Efficient routing
- ③ Communication b/w devices.

→ OBSERVATION

During the study of the college network infrastructure:—

- ① A centralized server room was used of network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-link POE switches were used in network cabinets.

```
PS C:\Users\tamke> netsh wlan
```

The following commands are available:

Commands in this context:

?	- Displays a list of commands.
add	- Adds a configuration entry to a table.
connect	- Connects to a wireless network.
delete	- Deletes a configuration entry from a table.
disconnect	- Disconnects from a wireless network.
dump	- Displays a configuration script.
export	- Saves WLAN profiles to XML files.
help	- Displays a list of commands.
IHV	- Commands for IHV logging.
refresh	- Refresh hosted network settings.
reportissues	- Generate WLAN smart trace report.
set	- Sets configuration information.
show	- Displays information.
start	- Start hosted network.
stop	- Stop hosted network.

To view help for a command, type the command, followed by a space, and then type ?.

```
PS C:\Users\tamke> |
```

```
PS C:\Users\tamke> netsh wlan show interface
```

There is 1 interface on the system:

Network shell commands need location permission to access WLAN information. Turn on Location services on the Location page in Privacy & security settings.

Here is the URI for the Location page in the Settings app:

```
ms-settings:privacy-location
```

To open the Location page in the Settings app, hold down the Ctrl key and select the link, or run the following command:

```
start ms-settings:privacy-location
```

Or, to open the Location page from the Run dialog box, press Windows logo key + R, and then copy and paste the URI above.

Function WlanQueryInterface returns error 5:

The requested operation requires elevation (Run as administrator).

```
PS C:\Users\tamke> |
```

```
PS C:\Users\tamke> ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : BN11.COM
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::9f48:5cb3:7530:e5f%14
IPv4 Address. . . . . : 10.45.243.239
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.45.243.54
```

```
PS C:\Users\tamke> |
```

```
PS C:\Users\tamke> getmac
```

Physical Address	Transport Name
A0-AD-9F-50-E2-4B	Media disconnected
58-02-05-D9-EB-CC	\Device\Tcpip_{B598807E-C761-4FCF-91F2-EC804FB6D44A}

```
PS C:\Users\tamke> |
```

→ Advantage of Existing Network?

- * Scalable architecture.
- * centralized management
- * Supports POE device.
- * Easy expansion of blocks & classroom.
- * support both wireless & wired.
- * Redundant link for failover.

CONCLUSION

The obsenced campus LAN follow a Hybrid hierarchical topology consist of Tree, Extended Star, Ring. The network uses managed POE switches, wireless access point, Ethernet port & centralized-server infrastructure to provide efficient wired & wireless communication across the campus.